

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy at Project Check-in.

Your abstract should be written after you finish the research and experimentation and should include:

- **Project number**, title and the full name of every team member
- Purpose of the project
- Procedure
- Observations/data/results
- Conclusions
- 250 words or less (to fit on one page)
- Use a black font size 10 to 12

Project A74: Fuzzing the Proofs: Designing a Coverage-guided Software to Reveal Vulnerabilities in Zero-Knowledge Proofs

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Zero-knowledge proofs (ZKPs) allow one party to prove the correctness of a computation without revealing any extraneous data. While the theory behind these systems is well understood, real implementations often rely on complex circuits where even the smallest mistakes can jeopardize security. The purpose of this project is to design a system that can detect correctness flaws in ZKP circuits and study how well such a detection mechanism scales as circuit vulnerabilities increase.

To address this problem, I developed zkSem, a testing framework for Circom-based ZKP circuits. zkSem combines static analysis, dynamic fuzzing, and metamorphic circuit transformations to expose constraint inconsistencies and semantic errors. Each test run of my software records timing information, failure signatures, and execution outcomes. Two benchmark suites were constructed: the first contains real-world circuits with known errors collected from existing ZKP implementations, while the second introduces controlled bug families where the number of injected bugs increases incrementally.

My results show that zkSem was able to successfully detect correctness issues across all known vulnerable circuits. Furthermore, static analysis identified the specific underlying structural issues such as missing range checks and unconstrained assignments, while dynamic tests revealed constraint failures in these circuits. In scalability experiments, detection latency remained stable and throughput did not significantly decrease as the number of bugs increased.

Together, these results suggest that zkSem can help developers identify subtle implementation errors in ZKP circuits and provide a foundation for improving and scaling the reliability of these cryptographic systems.